Campus Network Subnetting & IP Address Plan

Based on Network Topology & Cisco Multi-Area OSPF Configuration

1. Complete IP Address Calculation Table

Segment / Role	VLAN	Prefix	Network Addr	Subnet Mask	Wildcard	Usable Host Range	Broadcast	Usable Hosts
MLS-CSE Student	10	/21	192.168.0.0	255.255.248.0	0.0.7.255	192.168.0.1 – 192.168.7.254	192.168.7.255	2046
MLS-CSE Teacher	20	/22	192.168.24.0	255.255.252.0	0.0.3.255	192.168.24.1 – 192.168.27.254	192.168.27.255	1022
MLS-CSE Academic	30	/23	192.168.36.0	255.255.254.0	0.0.1.255	192.168.36.1 – 192.168.37.254	192.168.37.255	510
MLS-BBA Student	10	/21	192.168.8.0	255.255.248.0	0.0.7.255	192.168.8.1 – 192.168.15.254	192.168.15.255	2046
MLS-BBA Teacher	20	/22	192.168.28.0	255.255.252.0	0.0.3.255	192.168.28.1 – 192.168.31.254	192.168.31.255	1022
MLS-BBA Academic	30	/23	192.168.38.0	255.255.254.0	0.0.1.255	192.168.38.1 – 192.168.39.254	192.168.39.255	510
MLS-NFS Student	10	/21	192.168.16.0	255.255.248.0	0.0.7.255	192.168.16.1 – 192.168.23.254	192.168.23.255	2046
MLS-NFS Teacher	20	/22	192.168.32.0	255.255.252.0	0.0.3.255	192.168.32.1 – 192.168.35.254	192.168.35.255	1022
MLS-NFS Academic	30	/23	192.168.40.0	255.255.254.0	0.0.1.255	192.168.40.1 – 192.168.41.254	192.168.41.255	510
Core – MLS-CSE	N/A	/29	192.168.42.0	255.255.255.248	0.0.0.7	192.168.42.1 – 192.168.42.6	192.168.42.7	6
Core – MLS-BBA	N/A	/29	192.168.42.8	255.255.255.248	0.0.0.7	192.168.42.9 – 192.168.42.14	192.168.42.15	6
Core – MLS-NFS	N/A	/29	192.168.42.16	255.255.255.248	0.0.0.7	192.168.42.17 – 192.168.42.22	192.168.42.23	6
Core – ISP WAN	N/A	/24	103.133.254.0	255.255.255.0	0.0.0.255	103.133.254.1 – 103.133.254.254	103.133.254.255	254

2. Device Interface IP Assignments

Device	Interface	IP Address	Area	Description
Core Router	Gi0/0	103.133.254.2/24	N/A	Uplink to ISP
Core Router	Gi1/0	192.168.42.1/29	0	Downlink to CSE
Core Router	Gi2/0	192.168.42.9/29	1	Downlink to BBA
Core Router	Gi3/0	192.168.42.17/29	2	Downlink to NFS
MLS-CSE	Gi1/0/24	192.168.42.2/29	0	Uplink to Core
MLS-CSE	Vlan 10	192.168.0.1/21	0	Default Gateway
MLS-CSE	Vlan 20	192.168.24.1/22	0	Default Gateway
MLS-CSE	Vlan 30	192.168.36.1/23	0	Default Gateway
MLS-BBA	Gi1/0/24	192.168.42.10/29	1	Uplink to Core
MLS-BBA	Vlan 10	192.168.8.1/21	1	Default Gateway
MLS-BBA	Vlan 20	192.168.28.1/22	1	Default Gateway
MLS-BBA	Vlan 30	192.168.38.1/23	1	Default Gateway
MLS-NFS	Gi1/0/24	192.168.42.18/29	2	Uplink to Core
MLS-NFS	Vlan 10	192.168.16.1/21	2	Default Gateway
MLS-NFS	Vlan 20	192.168.32.1/22	2	Default Gateway
MLS-NFS	Vlan 30	192.168.40.1/23	2	Default Gateway
ISP Router	Gi0/0/0	103.133.254.1/24	N/A	Gateway to Internet
External	NIC	103.133.254.3/24	N/A	External Server

3. DHCP Pools & Infrastructure Mapping

Pool Name	Network	Default Router	DNS Server
CSE_STUDENT	192.168.0.0/21	192.168.0.1	103.133.254.2
CSE_TEACHER	192.168.24.0/22	192.168.24.1	103.133.254.2
CSE_ACADEMIC	192.168.36.0/23	192.168.36.1	103.133.254.2
BBA_STUDENT	192.168.8.0/21	192.168.8.1	103.133.254.2
BBA_TEACHER	192.168.28.0/22	192.168.28.1	103.133.254.2
BBA_ACADEMIC	192.168.38.0/23	192.168.38.1	103.133.254.2
NFS_STUDENT	192.168.16.0/21	192.168.16.1	103.133.254.2
NFS_TEACHER	192.168.32.0/22	192.168.32.1	103.133.254.2
NFS_ACADEMIC	192.168.40.0/23	192.168.40.1	103.133.254.2

4. OSPF Architecture

- **Core ID:** 1.2.3.0 (ABR)
- **MLS-CSE ID:** 1.1.1.1 (Area 0)
- **MLS-BBA ID:** 2.2.2.2 (Area 1)
- **MLS-NFS ID:** 3.3.3.3 (Area 2)
- **Default Route:** 0.0.0.0/0 via 103.133.254.1
- **Access Ports:** 1-6 (V10), 7-12 (V20), 13-18 (V30)
- **Uplink Port:** Port 24 (`no switchport`)

5. SSH & Remote Access

- **Protocol:** SSHv2 (Port 22), Domain: `campus.edu`
- **Key Modulus:** RSA 1024-bit (`crypto key generate`)
- **Credentials (Priv 15):** `admin`, `cisco`
- **Password:** `cisco` (for all users & enable secret)
- **VTY Lines:** 0–15 (`transport input ssh`, `login local`)
- **Console Line:** line con 0 (`password cisco`, `login`)
- **Mgmt IPs:** Core: 192.168.42.1, CSE: 192.168.42.2, BBA: 192.168.42.10, NFS: 192.168.42.18, ISP: 103.133.254.1